Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

EMBEDDED SYSTEMS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.

(2) Assume suitable data wherever necessary.

1. Recall what you understand by ARM Cortex as a RISC controller. 6(CO1)

2. Illustrate the programmer's model of Cortex M microcontroller with the help of processor mode, privilege level and registers of microcontroller. 6(CO2)

3. CPAC, what is it and where is it utilized in arm and why ? 6(CO1)

4. Interpret Race condition and its side effects in embedded systems. 6(CO3)

5. Organizing with neat diagram and example of your choice to generate a waveform, how will you utilize input capture in timers in cortex M for the said generation ? 6(CO3)

6. Discriminate ADC with polling as well as without polling. Give example of each in cortex M with Embedded C Code. 10(CO4,5)

7. Checking the use of a pulse width modulation in Cortex M microcontroller with an example program for a DC motor Interfaced at a GPIO pin. 10(CO4,5)

8. Construct a circuit such that a switch is connected to a digital input pin of Cortex M and a Seven Segment Display Common Anode is connected on a GPIO port. Write an embedded code that displays the updating count of the switch pressed on the Seven Segment Display. 10(CO4,5)